stmtikz package description

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v20250919

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These are the tikz definitions for `stmlatex`. It is build upon the tikz package.

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1. Usage - in the preamble

1.1. Load the whole `stmtikz` package

1.1.1. Description

This is an interface package which loads `tikz` and definitions commonly required throughout document creation.

By default the package loads

- `stmtikzbase.sty`
- `stmtikzlibraries.sty`
- `stmtikzstyles.sty`
- `stmtikzexternalization.sty`

See subsubsection 1.1.2 for options to change the default package behavior.

1.1.2. Options

Option *libraries* This is a boolean option. Expected values are either `true` or `false`. It controls whether to load the standard libraries commonly required.

```
\usepackage[libraries=true|false]{stmtikz}
```

`libraries=true` is the default. It is used in case `libraries=false` is not set explicitly.

Option *styles* This is a boolean option. Expected values are either `true` or `false`. It controls whether to load the predefined `tikz` styles.

```
\usepackage[styles=true|false]{stmtikz}
```

`styles=true` is the default. It is used in case `styles=false` is not set explicitly.

Option *externalization* This is a boolean option. Expected values are either `true` or `false`. It enables and disables the possibilities for the externalization of `tikzpictures`.

```
\usepackage[externalization=true|false]{stmtikz}
```

`externalization=true` is the default. It is used in case `externalization=false` is not set explicitly.

By default, global externalization is not set by the package. Thus it is required to use `\tikzexternalenable` before the first picture, which is to be created with this feature. See Option *globalexternalization* for a possibility to allow a global externalization of throughout the document.

Option *externalizationoutputfolder* This option expects a string input. Do not add a slash at the end of the string.

With this option it is possible to define a output folder for all externalized `tikzpicture`s in case Option *externalization* has the value `true`. The folder location is set relative to the directory of the main `tex`-file.

```
\usepackage[externalizationoutputfolder=$FOLDERNAME$]{stmtikz}
```

The default is `externalizationoutputfolder=ZZZ_TikZ`.

Option *globalexternalization* This is a boolean option. Expected values are either `true` or `false`.

By default externalization is not enabled for `tikzpicture`s globally, meaning automatically activated for each `tikzpicture`. It has to be activated explicitely in the document with `\tikzexternalenable`.

It is possible to control this behavior with

```
\usepackage[globalexternalization=true|false]{stmtikz}
```

`globalexternalization=false` is the default. It is used in case `globalexternalization=true` is not set explicitly.

Global externalization is active until the next `\tikzexternaldisable` in your document.

1.2. `stmtikzlibraries`

This package contains standard libraries commonly required in the creation of `tikzpicture`s.

1.3. `stmtikzstyles`

This package contains styles commonly required in the creation of `tikzpicture`s.

1.4. `stmtikzexternalization`

1.4.1. Description

This package enables the externalization of `tikzpicture`s.

It requires shell escape. Enabling shell escape is achieved by using the `-shell-escape` command line option when executing LaTeXor pdfLaTeX, e.g.

```
pdflatex -shell-escape stmtikz.tex
```

or with `arara`

```
arara: pdflatex: {shell: yes}
```

1.4.2. Options

Option *outputfolder* This option expects a string input. Do not add a slash at the end of the string.

With this option it is possible to define a output folder for all externalized `tikzpicture`s. The folder location is set relative to the directory of the main `tex`-file.

```
\usepackage[outputfolder=$FOLDERNAME$]{stmtikzexternalization}
```

The default is `outputfolder=ZZZ_TikZ`.

Option *globalexternalization* This is a boolean option. Expected values are either `true` or `false`.

By default externalization is not enabled for `tikzpicture`s globally, meaning automatically activated for each `tikzpicture`. It has to be activated explicitely in the document with `\tikzexternalenable`.

It is possible to control this behavior with

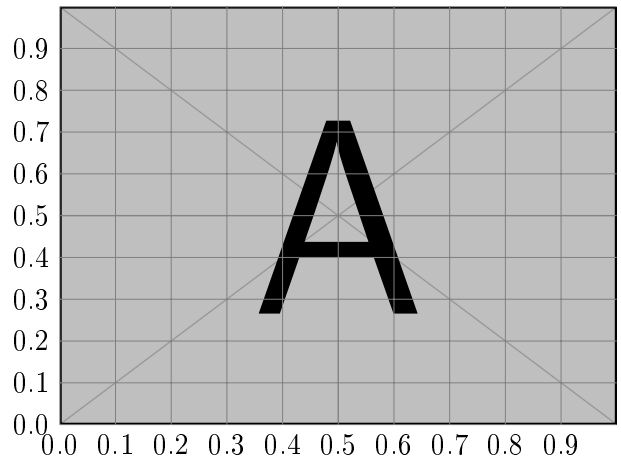
```
\usepackage[globalexternalization=true|false]{stmtikzexternalization}
```

`globalexternalization=false` is the default. It is used in case `globalexternalization=true` is not set explicitly.

Global externalization is active until the next `\tikzexternaldisable` in your document.

2. Externalization

This is a test for the externalization of a `tikzpicture`.



3. Known issues

misplaced `\printindex` error.

- There is an issue with the externalization of `tikzpictures` in case the `imakeidx` package is used.
- There are two possibilities:
  1. Load the `imakeidx` before any call to `stmtikzexternalization`

     ```
     \usepackage{imakeidx}
     \usepackage[externalization=true]{stmtikz}
     ```
  2. Load the package without externalization and load the feature after the call to `imakeidx`

     ```
     \usepackage[externalization=false]{stmtikz}
     \usepackage{imakeidx}
     \usepackage{stmtikzexternalization}
     ```

A. The code

A.1. stmtikz.sty

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Header                              %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%
% This is a interface to all stm tikz definitions
% Based upon the tikz package:
%   https://ctan.org/pkg/tikz
%
% Usage
%  - Preamble:
%    - \usepackage{stmtikz}
%
% Revisions: 2019-10-27 Martin Raedel <martin.raedel@dlr.de>
%                       Initial draft
%
% Contact:   Martin Raedel,  martin.raedel@dlr.de
%            DLR Lightweight Systems
%
%                                 __/|__
%                                /_/_/_/
%            www.dlr.de/sy/en      |/ DLR
%
% Copyright (C) 2019-... DLR Lightweight Systems
%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Content                             %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

% Declare that this style file requires at least LaTeX version 2e.
\NeedsTeXFormat{LaTeX2e}

% Provide the name of your page, the date it was last updated, and a
    comment about what it's used for
\ProvidesPackage{stmtikz}[2023/02/12 STMs custom LaTeX tikz definitions]

% ---------------------------
% Options
% ---------------------------

% For options
```

```
\@ifpackageloaded{kvoptions}{}{\RequirePackage{kvoptions}}%

% Option group
\SetupKeyvalOptions{%
  family=stmtikz,%
  prefix=stmtikz@,%
  setkeys=\kvsetkeys,%
}

% Libraries
\DeclareBoolOption[true]{libraries}

% Styles
\DeclareBoolOption[true]{styles}

% Externalization
\DeclareBoolOption[true]{externalization}

% Global externalization
\DeclareBoolOption[false]{globalexternalization}

% Output folder
\DeclareStringOption[ZZZ_TikZ]{externalizationoutputfolder}

% Process options
\ProcessKeyvalOptions{stmtikz}

% ----------------------------
% Modules
% ----------------------------

% Load the base package
\@ifpackageloaded{stmtikzbase}{}{%
  \RequirePackage{stmtikzbase}%
}%

% Libraries
\ifstmtikz@libraries
  \RequirePackage{stmtikzlibraries}
\fi

% Styles
\ifstmtikz@styles
  \RequirePackage{stmtikzstyles}
```

```
\fi

% Externalization
\ifstmtikz@externalization
  \@ifpackageloaded{stmtikzexternalization}{}{
    \RequirePackage[%
      outputfolder=\stmtikz@externalizationoutputfolder,
      global={\ifstmtikz@globalexternalization true\else false\fi}
    ]{stmtikzexternalization}
  }
\fi

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% That's it                         %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

% Finally, we'll use \endinput to indicate that LaTeX can stop reading
    this file. LaTeX will ignore anything after this line.
\endinput
```

A.2. stmtikzbase.sty

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Header                            %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%
% This is a interface to stm base tikz package
% Based upon the tikz package:
%   https://ctan.org/pkg/tikz
%
% Usage
%  - Preamble:
%    - \usepackage{stmtikzbase}
%
% Revisions: 2019-10-27 Martin Raedel <martin.raedel@dlr.de>
%                       Initial draft
%
% Contact:   Martin Raedel,  martin.raedel@dlr.de
%            DLR Lightweight Systems
%
%                                 __/|__
%                                /_/_/_/
%            www.dlr.de/sy/en      |/ DLR
%
```

```
% Copyright (C) 2019-... DLR Lightweight Systems
%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Content                           %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

% Declare that this style file requires at least LaTeX version 2e.
\NeedsTeXFormat{LaTeX2e}

% Provide the name of your page, the date it was last updated, and a comment about what it's used for
\ProvidesPackage{stmtikzbase}[2023/02/12 STMs custom LaTeX tikz definitions]

% ---------------------------
% Package
% ---------------------------

\@ifpackageloaded{tikz}{}{\RequirePackage{tikz}}%

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% That's it                         %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

% Finally, we'll use \endinput to indicate that LaTeX can stop reading this file. LaTeX will ignore anything after this line.
\endinput
```

A.3. stmtikzlibraries.sty

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Header                            %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%
% This file includes unit definitions.
% Based upon the tikz package:
%   https://ctan.org/pkg/tikz
%
% Usage
%  - Preamble:
%    - \usepackage{stmtikzlibraries}
%
% Revisions: 2019-10-27 Martin Raedel <martin.raedel@dlr.de>
%                       Initial draft
```

```
%
% Contact:   Martin Raedel,  martin.raedel@dlr.de
%            DLR Lightweight Systems
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%                                  __/|__
%                                 /_/_/_/
%            www.dlr.de/sy/en       |/ DLR
%
% Copyright (C) 2019-... DLR Lightweight Systems
%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Content                            %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

% Declare that this style file requires at least LaTeX version 2e.
\NeedsTeXFormat{LaTeX2e}

% Provide the name of your page, the date it was last updated, and a
    comment about what it's used for
\ProvidesPackage{stmtikzlibraries}[2023/02/12 STMs custom LaTeX tikz
    library definitions]

% ---------------------------
% Package
% ---------------------------

\@ifpackageloaded{tikz}{}{\RequirePackage{tikz}}%

% ---------------------------
% Libraries
% ---------------------------

\usetikzlibrary{arrows.meta}
\usetikzlibrary{backgrounds}
\usetikzlibrary{calc}
\usetikzlibrary{decorations.markings}
\usetikzlibrary{decorations.text}
\usetikzlibrary{fit}
\usetikzlibrary{intersections}
\usetikzlibrary{tikzmark}
\usetikzlibrary{trees}
\usetikzlibrary{matrix}
\usetikzlibrary{mindmap}
\usetikzlibrary{patterns}
```

```
\usetikzlibrary{positioning}
\usetikzlibrary{shadows}
\usetikzlibrary{shadows.blur}
\usetikzlibrary{shapes}
\usetikzlibrary{shapes.misc}
\usetikzlibrary{spy}

% ----------------------------
% Pgfmath
% ----------------------------

% Stuff for calc compatiability.
\let\real=\pgfmath@calc@real
\let\minof=\pgfmath@calc@minof
\let\maxof=\pgfmath@calc@maxof
\let\ratio=\pgfmath@calc@ratio
\let\widthof=\pgfmath@calc@widthof
\let\heightof=\pgfmath@calc@heightof
\let\depthof=\pgfmath@calc@depthof

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% That's it                         %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

% Finally, we'll use \endinput to indicate that LaTeX can stop reading
    this file. LaTeX will ignore anything after this line.
\endinput
```

A.4. stmtikzstyles.sty

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Header                            %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%
% This is a interface to all stm tikz definitions
% Based upon the tikz package:
%   https://ctan.org/pkg/tikz
%
% Usage
%  - Preamble:
%    - \usepackage{stmtikzstyles}
%
% Revisions: 2019-10-27 Martin Raedel <martin.raedel@dlr.de>
%                       Initial draft
```

```
%
% Contact:   Martin Raedel,  martin.raedel@dlr.de
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%
% Copyright (C) 2019-... DLR Lightweight Systems
%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Content                            %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

% Declare that this style file requires at least LaTeX version 2e.
\NeedsTeXFormat{LaTeX2e}

% Provide the name of your page, the date it was last updated, and a
    comment about what it's used for
\ProvidesPackage{stmtikzstyles}[2023/02/12 STMs custom LaTeX tikz style
    definitions]

% ----------------------------
% Package
% ----------------------------

\@ifpackageloaded{tikz}{}{\RequirePackage{tikz}}%
\@ifpackageloaded{stmcolors}{}{\RequirePackage{stmcolors}}%

% ----------------------------
% Pgfkeys
% ----------------------------

% From the pgf manual 2.10csv page 694:
%
%     It should be noted that all calculations must not exceed
    +-16383.99999 at any point, because the underlying computations rely
    on TeX dimensions. This means that many of the underlying computations
     are necessarily approximate and that in addition, are not very fast.
    TeX is, after all, a typesetting language and not ideally suited to
    relatively advanced mathematical operations. However, it is possible
    to change the computations as described in Section 76.
%
% From the TeX Book page 114:
```

```
%
%     16383.99998pt (TeX's largest dimen)
%
% In Notes On Programming in TeX Chirstian Feuersaenger pointed out
%
%     The \dimen registers perform their arithmetic's internally with 32
    bit scaled integers, so called scaled point with unit sp. It holds 1pt
     = 65536sp = 216sp. One of the 32 bits is used as sign. The total
    number range in pt is [-(2^30-1)/2^16,(2^30-1)/2^16 ]=[-16383.9998,
    +16383.9998]1.
%
%     1 Please note that this does not cover the complete range of a 32
    bit integer, I do not know why
% \pgfkeys{/pgf/fpu=true}                         % Allow pgf to plot
    values larger than 16383.9998
% \tikzset{fpu}                                     % Allow pgf to plot
    values larger than 16383.9998

\pgfkeys{/tikz/savenumber/.code 2 args={\global\edef#1{#2}}}

% ---------------------------
% Styles
% ---------------------------

% Help grid for external images
% Call inside scope with: \pic{stmimagegrid};
\tikzset{%
  stmimagegrid/.pic={%
   \draw[help lines,xstep=.1,ystep=.1] (0,0) grid (1,1);%
   \foreach \x in {0,1,...,9} {\node [anchor=north] at (\x/10,0) {0.\x};}
    %
   \foreach \y in {0,1,...,9} {\node [anchor=east]  at (0,\y/10) {0.\y};}
    %
  }%
}

% Default arrow tip
\tikzset{
  stmdefaultarrow/.style={                              % Define arrow style
    >=Stealth,                                  % >=latex
  }
}

% A cross for markings in plots
```

```
% https://tex.stackexchange.com/a/124064
\tikzset{%
  cross/.style={%
    cross out,%
    draw,%
    minimum size=2*(#1-\pgflinewidth),%
    inner sep=0pt,%
    outer sep=0pt,%
  }%
}%

% Beamer overlay feature
% https://tex.stackexchange.com/a/146991/44634
\tikzset{%
  invisible/.style={%
    opacity=0,%
    text opacity=0%
  },%
  visible on/.style={%
    alt={%
      #1{}{invisible}%
    }%
  },%
  alt/.code args={<#1>#2#3}{%
    \alt<#1>{\pgfkeysalso{#2}}{\pgfkeysalso{#3}} % \pgfkeysalso doesn't
    change the path
  },
}

\tikzset{%
  stmmarkuptext/.style={%
    text=stmmarkupcolor,%
  }%
}

% Markup style for rectangles on external figures
\tikzset{%
  stmrectangularmarkup/.style={%
   inner sep=0pt,% necessary for correct positioning of corners
   draw=stmmarkupcolor,%
   thick,%
  }%
}

```

```
% Equal space decoration markers along addplot path
% http://tex.stackexchange.com/a/232010/44634
\tikzset{
  nomorepostactions/.code={\let\tikz@postactions=\pgfutil@empty},
  stmmark/.style 2 args={decoration={markings,
    mark= between positions 0 and 1 step (1/11)*\pgfdecoratedpathlength
    with{%
        \tikzset{#2,every mark}\tikz@options
        \pgfuseplotmark{#1}%
      },
    },
    postaction={decorate},
    /pgfplots/legend image post style={
      mark=#1,
      #2,
      every path/.append style={nomorepostactions}
    },
  },
}

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% That's it                          %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

% Finally, we'll use \endinput to indicate that LaTeX can stop reading
    this file. LaTeX will ignore anything after this line.
\endinput
```

A.5. stmtikzexternalization.sty

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Header                             %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%
% This is a interface to tikz externalization definitions.
% All TikZ output is written to 'ZZZ_TikZ' folder.
% Based upon the tikz package:
%   https://ctan.org/pkg/tikz
%
% Usage
%  - Preamble:
%    - \usepackage{stmtikzexternalization}
%
% Revisions: 2019-10-27 Martin Raedel <martin.raedel@dlr.de>
```

```
%                       Initial draft
%
% Contact:   Martin Raedel,  martin.raedel@dlr.de
%            DLR Lightweight Systems
%
%                                 __/|__
%                                /_/_/_/
%            www.dlr.de/sy/en      |/ DLR
%
% Copyright (C) 2019-... DLR Lightweight Systems
%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Content                            %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

% Declare that this style file requires at least LaTeX version 2e.
\NeedsTeXFormat{LaTeX2e}

% Provide the name of your page, the date it was last updated, and a
    comment about what it's used for
\ProvidesPackage{stmtikzexternalization}[2023/02/12 STMs custom LaTeX
    tikz externalization definitions]

% ---------------------------
% Package
% ---------------------------

\@ifpackageloaded{tikz}{}{\RequirePackage{tikz}}%

% ---------------------------
% Options
% ---------------------------

% Options package
\@ifpackageloaded{kvoptions}{}{\RequirePackage{kvoptions}}%

% Option group
\SetupKeyvalOptions{%
  family=stmtikzexternalization,%
  prefix=stmtikzexternalization@,%
  setkeys=\kvsetkeys,%
}

% Externalization
```

```
\DeclareBoolOption[false]{global}

% Output folder
\DeclareStringOption[ZZZ_TikZ]{outputfolder}

% Process options
\ProcessKeyvalOptions{stmtikzexternalization}

% ----------------------------
% Libraries
% ----------------------------

\usetikzlibrary{external}

% ----------------------------
% Settings
% ----------------------------

\tikzexternalize[%
  mode=convert with system call,%
  shell escape=-enable-write18,%  % Use for MiKTeX
]

% Output folder
\tikzsetexternalprefix{\stmtikzexternalization@outputfolder/}

% Set
\tikzset{%
  external/system call={%
    pdflatex \tikzexternalcheckshellescape -halt-on-error %
    -interaction=batchmode -jobname "\image" "\texsource" %&& %
    %convert -density 600 -transparent white "\image.pdf" "\image.png" %
    for ImageMagick versions <7
    %magick  -density 600 -transparent white "\image.pdf" "\image.png" %
    for ImageMagick versions >= 7
  }
}

% By default, do not allow global externalization
\ifstmtikzexternalization@global
  \tikzexternalenable
\else
  \tikzexternaldisable
\fi
```

```
99
100 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
101 % That's it                          %
102 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
103
104 % Finally, we'll use \endinput to indicate that LaTeX can stop reading
        this file. LaTeX will ignore anything after this line.
105 \endinput
```